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θ-Fe_3C dominated Fe@C core–shell catalysts for Fischer-Tropsch synthesis: Roles of θ-Fe_3C and carbon shell

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ABSTRACT

θ-Fe_3C formation is usually considered as a deactivation factor in Fischer-Tropsch synthesis (FTS). Herein, θ-Fe_3C dominated Fe@C core–shell catalysts were fabricated by pyrolysis of MIL-101(Fe) in ethyne for studying the roles of θ-Fe_3C and carbon shell in FTS. In the evolution of MIL-101(Fe) into Fe@C catalysts, ethyne decomposition promotes the formation of θ-Fe_3C and carbon shells around nanoparticles. By modulating the pyrolysis temperature, θ-Fe_3C and/or α-Fe nanoparticles coated with different carbon shells (amorphous carbon, graphene or graphite shells) are acquired. By exploring the relationship between phase composition, structure of catalysts and FTS performance, it is found that θ-Fe_3C is confirmed to be FTS active, which exhibits superior C—C chain growth ability, i.e., higher C_{5+} selectivity and lower CH_4 selectivity. Meanwhile, graphene shell inhibits iron nanoparticles aggregation and stabilizes the catalysts.

1. Introduction

Fast energy consumption and limited crude oil reserves have driven worldwide research on petroleum substitutes [1,2]. Fischer-Tropsch synthesis is an attractive process for the production of chemicals and clean liquid fuels from carbon resources, such as coal, natural gas, and biomass. Iron-based catalysts have been industrially applied in FTS owning to their low cost, adjustable selectivity, and relatively high activity over a wide range temperature (220–350 °C) [3]. However, the iron-based catalysts are susceptible to deactivation as a result of phase transformation, sintering, and carbon deposition during reaction [4]. Therefore, controlling the active phase, maintaining the metal dispersion, and illustrating the roles of carbon are extremely important for the optimization of FTS performance of iron-based catalysts.

Carbon materials are usually used as catalyst supports due to their inert surface properties and huge surface area. For iron-based FTS catalysts, carbon supports have special merits for improving iron carbide content and dispersion, and avoiding strong metal-support interactions. However, the role of carbon in FTS reaction remains on debate. For example, carbon nanotube encapsulated iron catalysts prepared by conventional impregnation method showed an intriguing confinement effect of carbon nanotube, and exhibited an enhanced activity and increased long-chain hydrocarbon selectivity [5]. Recently, metal organic frameworks (MOFs) derived Fe@C catalysts also exhibited unprecedented activities (ca. 0.44 $mmol_{CO}/g_{Fe}/s$) in high temperature FTS reaction [4]. In contrast, carbon depositions are considered to be one of the reasons for catalyst deactivation. Sun et al reported that Co@C hybrids prepared by directly pyrolyzing ZIF-67 showed a poor FTS activity [6]. The inferior FTS performances of Co@C catalysts were ascribed to the inaccessibility of most cobalt nanoparticles, which were completely encapsulated by graphite shells. In principal, the carbon shell around particles could occupy the active sites, block the reactants/products transport channels, and decrease the catalytic activity. The inconsistent conclusions on carbon shell might be caused by the different layers and carbon species. Is it possible to fabricate different carbon shell (amorphous shell, graphene shell and graphite shell) in a controllable way to explore their effects on the catalytic performance? Up to now, nearly no work concerns the role of different carbon shell in iron catalyzed FTS reactions.

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ε′-$Fe_{2.2}C$, χ-Fe_5C_2, and θ-Fe_3C are the most common iron carbide phases occurred in FTS [7,8]. ε′-$Fe_{2.2}C$ and χ-Fe_5C_2 have been considered definitely to be active [7], while the role of θ-Fe_3C in FTS remains controversial. It was reported that θ-Fe_3C appeared in unsupported iron catalyst during the deactivation process, which acted as an inactive phase of FTS [9]. Sergio Herranz et al. also reported that θ-Fe_3C was less active compared with χ-Fe_5C_2 in FTS reactions [10]. However, Liu et al reported that a relatively small amount of θ-Fe_3C modified by manganese exhibited a comparable FTS activity and an increased light olefins (C_2–C_4) selectivity compared with χ-Fe_5C_2 [11]. In addition, theoretical calculation also indicated that Fe_3C (001) is relatively less active toward CH_4 formation [12]. In summary, θ-Fe_3C is a controversial phase in FTS reactions and it is of significance to clarify the role of θ-Fe_3C.

In this work, the roles of θ-Fe_3C and different carbon shell of θ-Fe_3C dominated Fe@C core–shell catalysts in FTS reaction were investigated. The catalysts were prepared by controlled pyrolysis of MIL-101(Fe) in ethyne. Different θ-Fe_3C contents and carbon shells (amorphous carbon, graphene shell, and graphite shell) in Fe@C catalysts were fabricated by controlling the pyrolysis temperature. For comparison, a Fe@C-600-H_2 catalyst was synthesized by reducing Fe@C-600 in H_2 and an unsupported-Fe catalyst was prepared by co-precipitation. It is well-known that alkali promoter such as potassium is essential component for iron-based FTS catalysts. However, alkalis have strong promotional effects on CO adsorption/dissociation, C—C coupling, carburization, carbon deposition and water-gas-shift reaction. The strong promotion of alkalis would hide the intrinsic roles of θ-Fe_3C and carbon shells. Therefore, we deliberately prepared the alkali-free Fe@C catalysts to avoid the interference of alkalis. The iron phase, compositions, and carbon structures of catalysts are characterized by X-ray diffraction (XRD), Mössbauer spectroscopy (MES), transmission electron microscopy (TEM) and Raman spectroscopy. This paper reveals that θ-Fe_3C possesses superior selectivity towards long-chain hydrocarbons and the graphene shell has excellent stabilization effect on the active phase of catalysts in FTS reactions.

2. Experimental section

2.1. Catalysts preparation

MIL-101(Fe) was prepared by hydrothermal method. In a typical experimental procedure, 166 mg terephthalic acid (1 mmol), 606 mg $Fe(NO_3)_3 \cdot 9H_2O$ (1.5 mmol), 5 ml nitric acid (0.074 mmol) and 70 ml N'N-dimethylformamide (DMF) (0.9 mmol) were mixed with stirring. After stirring for 2 h, the homogeneous solution was transferred to a Teflon-lined steel autoclave (100 ml) which was then placed in an oven at 120 °C for 48 h. The products were washed with DMF and ethanol three times and then dried at 80 °C overnight.

MIL-101(Fe) was pyrolyzed in a flow of 2% C_2H_2/helium (50 ml/min) in a tube furnace (Thermal scientific). The products were pyrolyzed at 500, 600 and 700 °C for 1 h with a heating rate of 5 °C/min, respectively. The synthesized catalysts are denoted as 'Fe@C-x', with x representing the pyrolysis temperature. For comparison, Fe@C-600 was reduced in hydrogen (50 ml/min) at 400 °C for 24 h, named as Fe@C-600-H_2. The precursor of unsupported-Fe catalyst was prepared in a continuous co-precipitation process. The flowing aqueous solution containing $Fe(NO_3)_3$ (1 mol/L) and ammonia solution (3 mol/L) were added dropwise into a beaker of 100 ml under stirring at 80 ± 1 °C and pH 8.2–8.5. After aging for 2 h, the precipitate was washed and filtered with distilled water several times and dried for 12 h at 120 °C [13]. The sample was calcined for 5 h at 500 °C in static air and then for 24 h at 350 °C in H_2 (100 ml/min), named as unsupported-Fe. After pyrolysis, each catalyst was divided into two parts in a glovebox filled with N_2. One part was sealed with ethanol for characterization and the other part was loaded in reaction tube for FTS.

2.2. Catalysts characterization

The X-ray diffraction (XRD) patterns were recorded with an advanced D8 powder diffractometer (Bruker, Germany) using a Co Kα radiation (γ = 0.179 nm) operated at 35 kV and 40 mA. Measurements were carried out with a scan step of 0.03°, in the range of 2θ = 1–30° for MIL-101(Fe) and 2θ = 10–90° for catalysts as-prepared.

Mössbauer spectroscopy (MES) measurements were carried out in an MR-351 constant-acceleration Mössbauer spectrometer (FAST, Germany) driven with a triangular reference signal at room temperature. Data analysis was performed using the MossWinn 4.0 software package. The velocity was calibrated by α-Fe foil, and the Isomer Shift (IS) value was referenced to α-Fe.

N_2 adsorption measurements were carried out at −196 °C on a Micromeritics ASAP 2420 analyzer. Before analysis, MIL-101(Fe) was evacuated at 160 °C in vacuum for 12 h, and catalysts as-prepared were evacuated at 300 °C in vacuum for 12 h. The Raman spectra were recorded on a Lab Ram HR800 (Horiba Jobin-Yvon), using the charge-coupled device as a detector with 532 nm laser excitation. The concentration of metal in the catalysts was measured by inductively coupled plasma-optical emission spectrometry (ICP-OES, USA). Thermogravimetric analyses (TG) were conducted on a METTLER TOLEDO TGA-DSC 2 thermogravimetric analyzer. The samples were heated to 1000 °C at a rate of 10 °C/min in a flow of 60 ml/min.

Scanning electron microscopy (SEM) was performed on FEI Quanta 499 FEG equipment. Transmission electron microscopy (TEM) was performed on the FEI Talos 200A electron microscope operating at an accelerating voltage of 200 kV.

2.3. FTS test

FTS test was performed on a fixed bed reactor with an isothermal bed length of 5 cm. 0.5 g catalysts (40–60 mesh) diluted with 2.0 g of SiC particles (40–60 mesh) were loaded in the isothermal zone of the reactor. The remaining volume of the reactor was filled with SiC particles. The FTS reaction was carried out in a flow of syngas (H_2/CO = 1.0) at 300 °C, 2.0 MPa and a weight hourly space velocity (WHSV) of 8000 ml/g/h for MOFs derived catalysts and 3000 ml/g/h for unsupported-Fe. A detailed description of the reactor and product analysis systems has been presented elsewhere [14].

The tail gas was analyzed online by gas chromatographs (Model 6890 N and 4890D, Agilent, USA) with TCD and FID detectors. The water, oil and wax in the product were collected in a hot trap (160 °C) and a cold trap (0 °C) over 24 h for calculation of the mass balance and sampled for analysis. After reaction, the two ends of the reaction tube are blocked with sealing film when there is reaction gas. The reaction tube was immediately transferred to glove box to get the used-catalysts and soak them in ethanol. The catalytic activity was expressed in terms of iron time yield (FTY). The FTY was defined as moles of CO per gram of iron per second. The chain-growth probability value (α) is calculated from the slope of $\ln(W_n/n)$ (W_n: mass fraction of hydrocarbon, n:hydrocarbon chain length) as a function of n [15].

3. Results and discussion

3.1. Phase identification of catalysts

The crystalline structure of catalysts was characterized by XRD as shown in Fig. 1. The XRD pattern of Fe@C-500 exhibits the characteristic diffraction line of FeO (JCPDS 06-0615). The diffraction line in 50-55° (2θ) is too weak to determine the phase and further characterization is needed. The pattern of Fe@C-600 shows diffraction line of θ-Fe_3C (JCPDS 35-0772), Graphite-carbon (JCPDS 41-1487), and α-Fe (JCPDS 06–0696). In Fe@C-700, diffraction line corresponding to θ-Fe_3C and Graphite-carbon is observed. The diffraction line of Fe@C-700 becomes narrower and sharper, indicating a larger crystalline size and a higher graphitization degree of the carbon support. In Fe@C-600-H_2, the diffraction line is attributed to α-Fe and Graphite-carbon. The XRD pattern of unsupported-Fe shows diffraction line of α-Fe only. It can be seen that a higher reduction and carburization degree can be achieved in C_2H_2. C_2H_2 plays a crucial role in facilitating the formation of θ-Fe_3C.

The iron phase compositions and contents of catalysts were further characterized by MES analysis. Fig. 2 shows the Mössbauer spectra of catalysts and Table 1 exhibits the fitted parameters. The spectra are results of the superposition of sextets and/or doublets. The sextets with hyperfine magnetic field (Hhf) values of 208 and 206 kOe are attributed to the crystallographic sites in θ-Fe_3C. The structure of θ-Fe_3C includes two types of iron atoms: Fe (I) with twelve neighboring iron atoms and Fe (Ⅱ) with eleven neighboring iron atoms. The spectrum of Fe@C-500 is fitted with sextets attributing to θ-Fe_3C and doublets corresponding to Fe^{2+}, spm Fe^0 (superparamagnetic), and FeO. Fe@C-600 is composed of θ-Fe_3C, α-Fe, Fe^{2+}, and spmFe^0. Similarly, θ-Fe_3C, Fe^{2+}, and spmFe^0 are identified in Fe@C-700. The content of θ-Fe_3C in Fe@C-500, Fe@C-600, and Fe@C-700 is 68.2%, 87.2% and 95.5%, respectively. The increasing degree of reduction and carburization is consistent with XRD results. The spectrum of Fe@C-600-H_2 and unsupported Fe is corresponding to α-Fe. By reducing Fe@C-600 in H_2, θ-Fe_3C can be completely reduced to α-Fe. In addition, there is no other iron carbides phase (ε′-$Fe_{2.2}C$ and χ-Fe_5C_2) in these catalysts as-prepared.

According to the MES results, among the iron-containing phases, cementite (θ-Fe_3C) is the major phase in Fe@C-x catalysts (Fe@C-500, Fe@C-600 and Fe@C-700). The preparation of θ-Fe_3C has been previously reported. For example, De Smit et al synthesized θ-Fe_3C by exposing α-Fe_2O_3 to a flow of 10 ml/min CO at 450 °C, and reported that relatively high temperature led to the preferential formation of θ-Fe_3C [16]. Besides, Wezendonk et al reported that θ-Fe_3C was the predominant phase by pyrolysis of Fe-BTC at 900 °C in nitrogen atmosphere, and suggested that this was the result of lattice matching of the carbide with the iron matrix or the contributions of lattice vibration and anomalous magnetic ordering [17]. The generation of carbide phases can be modulated by controlling carbon potential (μ_c), which is influenced by starting materials, atmosphere, and temperature, etc. [13,16]. Then the synthesis strategy of θ-Fe_3C in this work is discussed in detail. In this work, the hydrogen and carbon decomposed from ethyne play a dominant role in promoting the reduction and carburization of iron species, and therefore large amount of θ-Fe_3C was produced. In addition, in Fe@C-x catalysts, the content of θ-Fe_3C increased from 68.2% in Fe@C-500 to 95.5% in Fe@C-700, which is proportional to the pyrolysis temperature. The TG curves of MIL-101(Fe) heated in inert gas as well as ethyne are shown in Fig. S1d. The two curves exhibit a similar weight loss trend before 300 °C. The weight loss of MIL-101(Fe) in ethyne is slightly lower than that in inert atmosphere between 300 °C and 500 °C. This manifests the carbon deposition as a result of ethyne decomposition. As the temperature increases to 500 °C, the weight loss of MOFs in ethyne is about 10% less than that in inert gas, denoting that the decomposition of ethyne is more intense. The further decomposition of ethyne in higher temperature promotes the formation of more θ-Fe_3C. Pyrolysis of MIL-101(Fe) in ethyne is proved effective to produce θ-Fe_3C since θ-Fe_3C is the dominant phase in Fe@C-x catalysts.

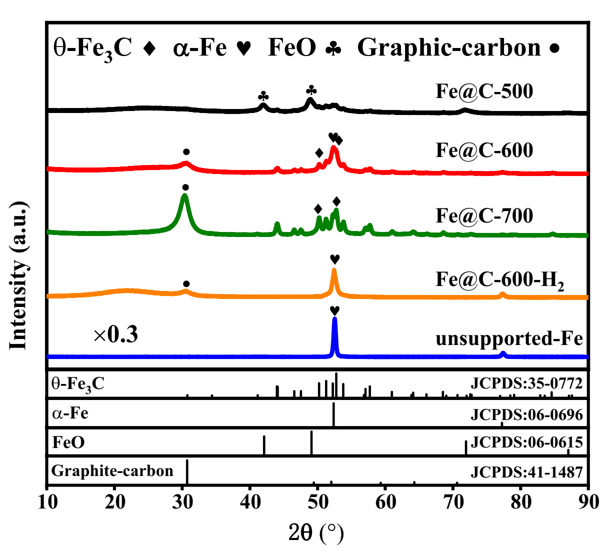

Fig. 1. XRD patterns of the catalysts.

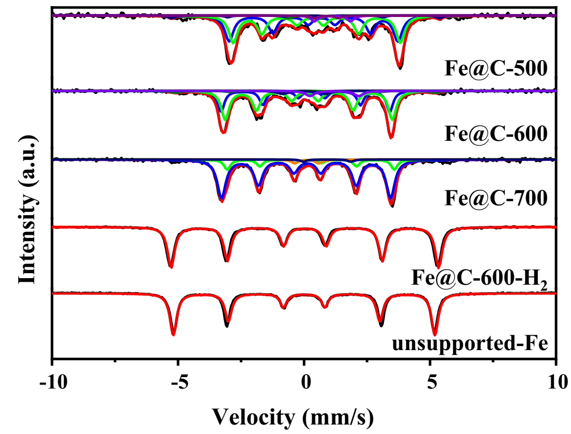

Fig. 2. Mössbauer spectra of the catalysts.

3.2. Textural and structure properties of catalysts

The textural properties and iron contents of the MIL-101(Fe) and catalysts are shown in Table 2. In contrast with MIL-101(Fe) (S_{BET} = 1632 m^2/g, V_{tol} = 0.83 cm^3/g), the S_{BET} and V_P of all MOFs derived catalysts as-prepared decreased to 135–291 m^2/g and 0.23–0.28 cm^3/g, respectively, which is due to the structure destruction of MIL-101(Fe). The unsupported-Fe catalyst showed very low specific surface area of 15.9 m^2/g. Compared with the iron oxide nanocrystals embedded in carbonaceous spheres using glucose and iron nitrate as starting materials [18], the MOFs derived catalysts have higher specific surface area as a result of the abundant pore structure of MOFs. As the pyrolysis temperature increases from 500 °C to 700 °C, the iron contents of Fe@C-x catalysts increase from 35.4% to 44.1%. From the TG curve in ethyne (Fig. S1d), MIL-101(Fe) shows a constant weight loss from 500 °C to 700 °C. The slightly increased iron content may due to the further ligands decomposition, which is demonstrated by the

Table 1
Mössbauer parameters of the catalysts.

Catalysts	IS (mm/s)	QS (mm/s)	Hhf (kOe)	Phase	Area (%)
Fe@C-500	0.2	0.12	208	θ-Fe_3C(I)	36.9
	0.5	−0.21	206	θ-Fe_3C(II)	31.3
	0	−2	–	spm Fe^0	2.2
	0.8	−0.92	–	Fe^{2+}	3.1
	1.0	1.44	–	FeO	26.5
Fe@C-600	0.05	−0.23	208	θ-Fe_3C(I)	45.6
	0	0.15	206	θ-Fe_3C(II)	41.6
	0	−1.66	–	spm Fe^0	2.1
	1.17	1.98	–	Fe^{2+}	3.9
	0.05	–	320	α-Fe	6.8
Fe@C-700	0.1	−0.05	208	θ-Fe_3C(I)	53.2
	0.21	0.1	206	θ-Fe_3C(II)	42.3
	0.1	−0.94	–	spm Fe^0	2.9
	0.92	1.88	–	Fe^{2+}	1.6
Fe@C-600-H_2	0.06	–	330	α-Fe	100
Unsupported-Fe	0	–	328	α-Fe	100

Table 2
Textural properties and Fe loadings of MIL-101(Fe) and catalysts.

Catalysts	S_{BET} (m^2/g)	V_p (cm^3/g)	Mean pore size (nm)	W_{Fe} (%)
MIL-101(Fe)	1632	0.83	1.3	18.4
Fe@C-500	135	0.23	3.3	35.4
Fe@C-600	280	0.27	3.8	36.5
Fe@C-700	165	0.28	3.8	44.1
Fe@C-600-H_2	291	0.28	3.7	38.2
Unsupported-Fe	15.9	0.04	–	100

continuous weight loss of MIL-101(Fe) in inert gas between 500 °C and 700 °C. The iron content of Fe@C-600-H_2 is 38.2%.

TEM was performed to analyze the structure of catalysts. TEM images of catalysts in Fig. 3a–d show the uniform distribution of darker nanoparticles encapsulated in a matrix with lower contrast. The elemental mapping of catalysts shown in Figs. S4–S7 confirms the coexistence of iron, carbon, and oxygen in catalysts. The oxygen from ligands decomposition exists in iron oxide and/or O-functionalization of carbon matrix [19]. The nanoparticles are related to iron while the matrix is ascribed to carbon. It can be observed that the area with a carbon signal overlapped with the profile of the catalysts. The iron and carbon mappings show that the intense signals of iron correspond to the spaces with lower carbon content, indicating the encapsulation of iron species in the carbon matrix of the catalysts. The metal ions or metallic clusters and organic ligands alternate periodically in MOFs structures. During the pyrolysis, the MOFs structure collapses accompanying the conversion of Fe clusters and organic ligands to nanoparticles and carbon matrix, respectively [20]. Therefore, nanoparticles are uniformly embedded in carbon matrix derived from the pyrolysis of MOFs. The TEM image of unsupported-Fe in Fig. 3e shows the relatively uniform particles without support. As shown in the insert graph of Fig. 3f–j, the lattice spacing (2.38 Å, 2.01 Å, 1.97 Å and 2.03 Å,) of the nanoparticles are consistent with the spacing of the Fe_3C (121), Fe_3C (031) Fe_3C (112), and Fe (110) crystal planes, respectively. The particle size distributions are shown in Fig. S8a–e. The nanoparticles display a narrow size distribution and have an approximate average size of 8.7 nm, 5.9 nm, 40 nm,

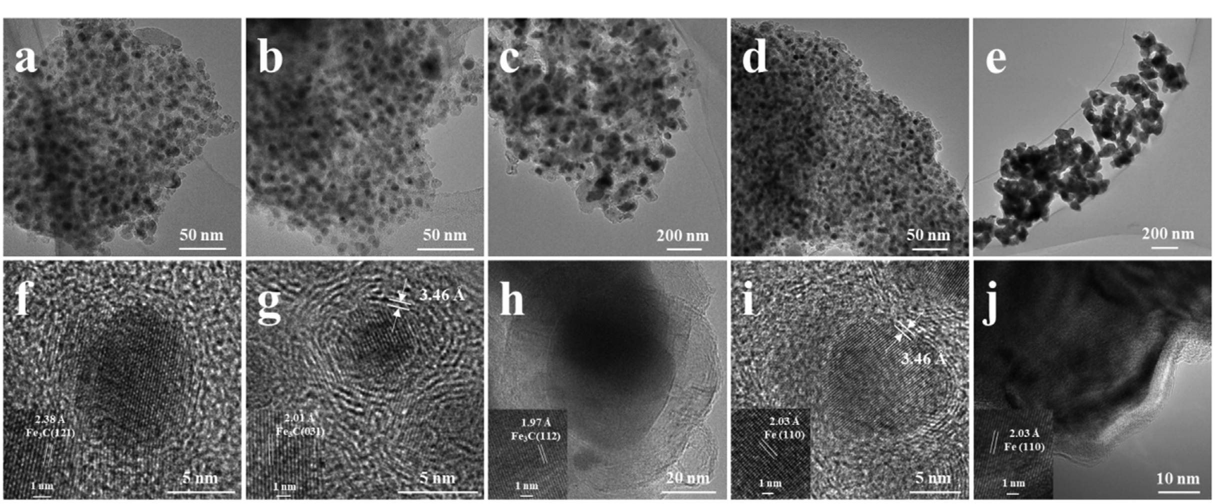

Fig. 3. TEM images (a–e) and HRTEM images (f–j). Fe@C-500: (a) and (f); Fe@C-600: (b) and (g); Fe@C-700: (c) (h); Fe@C-600-H_2: (d) and (i); unsupported-Fe: (e) and (j).

6.5 nm, and 63 nm in Fe@C-500, Fe@C-600, Fe@C-700, Fe@C-600-H_2 and unsupported-Fe, respectively. Among the Fe@C-x catalysts, the minimum nanoparticles are obtained in Fe@C-600 in this work. It is different from the general knowledge that small nanoparticles are thermally unstable and prone to aggregating into larger particles with increasing temperature. The influence of pyrolysis temperature on particle size needs further explanation by combining the analysis on carbon shell.

More information about the carbon shell is given in HRTEM (Fig. 3f–i). As shown in Fig. 3f, amorphous carbon around the particle is exhibited in Fe@C-500. In Fe@C-600 and Fe@C-600-H_2, a concentric graphene shell of 3–5 layers is formed in Fig. 3g and i. The reduction treatment in H_2 reduces the phase from iron carbide to α-Fe without obviously destroying the carbon shell structure. The interplanar spacing of graphene shell is 0.346 nm, close to the ideal graphite interplanar spacing (0.335 nm) [14]. In the HRTEM of Fe@C-700 (Fig. 3h), the layers of graphite shell increased dramatically, consistent with the XRD pattern. The graphite shell is well-oriented and densely stacked. In the MOFs derived catalysts, nanoparticles are packed in carbon shell, forming a core–shell structure. To further examine the carbon shell structure, the catalysts were leached with 0.5 M aqueous solution of hydrochloric acid. Fig. S8f–i exhibits the HRTEM images of the collected suspended solids after ultrasonic treatment for 3 days. Amorphous carbon cavities are shown in Fe@C-500. Graphene shell cavities with dislocation and holes are exhibited in Fe@C-600 and Fe@C-600-H_2. The graphite shell cavity is shown in Fe@C-700. The size of cavities matches well with that of the particles in these catalysts.

Generally, during the pyrolysis of MOFs in inert atmosphere, amorphous carbon around the particles is formed and it can be transformed to graphite at 2000–3000 °C without catalyst [4,21]. The catalysts (Fe, Co, Ni) play an effectively catalytic role in reducing the synthesis temperature of graphite. In this work, ethyne decomposition releases carbon fragments. MIL-101(Fe) has high pore volumes, which allows the carbon fragments to penetrate through the pores. The growth of the carbon shell that occurs at the nanoparticle surface can be attributed to the catalytic effect of iron species and the orderly combination of carbon fragments [20]. The formation of graphene shell can effectively restrict the further growth of particles, thus forming smaller particles. The confinement effect of graphene shell can be confirmed by the fact that the particle size of Fe@C-600 is smaller than that of Fe@C-500, although the pyrolysis temperature is increased. In addition, the more intense decomposition of ethyne at higher temperature produces more carbon species which promote the dispersion of iron at 600 °C. When the temperature rises to 700 °C, the aggregation of particles dominates and the average size of the particles reaches 40 nm. Through the above elucidation, it is concluded that the carbon shell and particle size of core–shell structure of catalysts is modulated by pyrolysis temperature of MOFs in ethyne.

Raman spectroscopy is an effective and non-destructive technique to characterize the structure of carbon materials such as the defects, disordered structures, graphitization degree, and the layers of grapheme [22,23]. The Raman spectra are shown in Fig. 4 and the fitting parameters are exhibited in Table S1. In the range of 1000–2000 cm^{-1}, the Raman spectra of Fe@C-500 show two bands at 1346.7 and 1584.8 cm^{-1}, respectively. Four peaks at 1000–2000 cm^{-1} are fitted by Lorentzian curves in the Raman spectra of Fe@C-600 and Fe@C-700 and Fe@C-600-H_2, respectively. The band at 1330–1350 cm^{-1} arises from the breathing mode of A_{1g} symmetry of graphite carbon, named D band. This mode is forbidden in perfect graphite and only becomes active in the presence of disorder [24,25]. The band at 1580–1600 cm^{-1} is attributed to the in-plane bond-stretching motion of C sp^2 atoms, named G band [26]. The value of the D/G ratio (peak height ratio) can be used to evaluate the graphitization degree of carbon materials. The I_D/I_G ratio of Fe@C-500 and Fe@C-700 is 1.10 and 0.91, respectively. Fe@C-600 and Fe@C-600-H_2 show lower I_D/I_G ratio (0.79 and 0.78), implying the higher graphitization degree. In Fe@C-600, two bands exist at 1209 and 1515 cm^{-1} in addition to the G and D peaks. As for Fe@C-700, the two bands at 1236 and 1488 cm^{-1} are observed. As reported in literature, the band at around 1200 cm^{-1} is ascribed to the vibration of carbon atoms in straight chain hydrocarbons with more than seven carbon atoms, while the peak centered at 1500 cm^{-1} can be attributed to semicircle stretching of carbon atoms in single aromatic rings or fused aromatic rings [27].

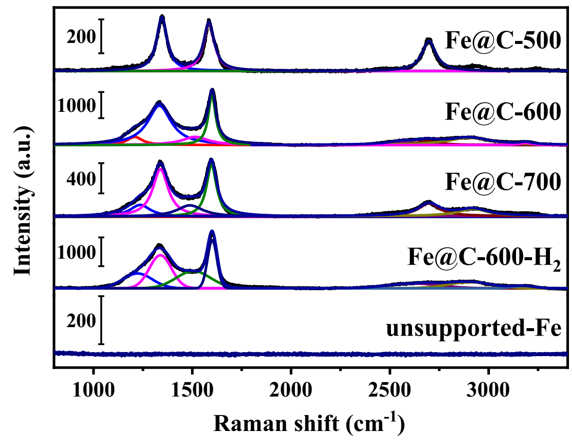

Fig. 4. Raman spectra of catalysts.

In 2500–3200 cm^{-1}, there are 2D bands originate from the second-order vibration caused by the scattering of phonons at the zone boundary [28]. The 2D bands are highly layer dependent and sensitive to the number of graphene layers [29]. In the monolayer graphene, a single, sharp 2D peak can be observed in Raman spectrum and the intensity of the 2D peak is four times that of the G peak. In the Raman spectrum of 3–5 layers of graphene, four peaks is exhibited, which is caused by vibration of 4–6 phonons with slightly different frequencies. As the difference of phonons vibration frequency decreases, the four peaks merge and eventually appear as two peaks ($2D_1$ and $2D_2$) in bulk graphite [30]. There is a weak and single peak at 2700 cm^{-1} in Fe@C-500, combined with XRD and TEM results, representing the presence of amorphous carbon. The broad band in Fe@C-600 and Fe@C-600-H_2 can be fitted to four peaks, implying the existence of 3–5 layers graphene shell. The Raman spectrum is fitted to two peaks in Fe@C-700, representing the formation of graphite shell. In 800–3400 cm^{-1}, there is no signal in the Raman spectra of unsupported-Fe. In this work, amorphous carbon, graphene shell, and graphite shell are formed, which is consistent with the HRTEM images. With the pyrolysis temperature increasing, the graphitization degree of carbon shells improved.

During the MIL-101(Fe) pyrolysis in ethyne, MIL-101(Fe) decomposes into carbon and iron species, and ethyne also splits into hydrogen and carbon species [14]. The diffusing and coalescing interaction between the iron and carbon fragments leads to the formation of α-Fe, θ-Fe_3C, and carbon shell under the reduction and carburization effect of ethyne [31]. Ethyne decomposes violently with increasing temperature, promoting the reduction and carburization of iron species. Therefore, the content of θ-Fe_3C is proportional to the pyrolysis temperature. The carbon fragments are condensed into carbon shells around the particles with the catalysis of iron. Amorphous carbon, graphene shell, and graphite shell is produced from 500 °C to 700 °C with higher graphitization degree. Among the carbon shell, the graphene shell constructs a physical barrier to inhibit the aggregation of iron nanoparticles.

3.3. FTS performances of catalysts

The FTS catalytic performances of catalysts were tested in conditions of 300 °C, 2.0 MPa, H_2/CO = 1.0. The H_2/CO ratio of 1.0 is chosen to represent coal or biomass derived syngas [1,32]. The CO conversions as a function of time on stream are presented in Fig. 5 and the FTS results are summarized in Table 3. All the mass balance of reactions lies in 95.5%–97.1%. Since the catalysts have different iron loadings, iron time yield (FTY) is used to compare the activities of catalysts. After 24 h on stream, the FTY values of the Fe@C-x catalysts are relatively high as 161–176 $\mu mol_{CO} \cdot g_{Fe}^{-1} s^{-1}$, a value better than the reported 150 $\mu mol_{CO} \cdot g_{Fe}^{-1} s^{-1}$ obtained from MOFs derived $Fe_3O_4@Fe_5C_2$ nanoparticles [32]. Additionally, Fe@C-600-H_2 exhibits even higher activity with CO conversion reaching 78.7% and the FTY value being 203.0 $\mu mol_{CO} \cdot g_{Fe}^{-1} s^{-1}$ at WGSV of 8000 ml/g/h. The unsupported-Fe catalyst shows lower CO conversion of 75.3% and FTY value of 39.5 $\mu mol_{CO} \cdot g_{Fe}^{-1} s^{-1}$ at WGSV of 3000 ml/g/h. Furthermore, the product distribution of all the catalysts is shown in Table 3. At 24 h, the Fe@C-x catalysts present identical yet satisfactory selectivity performance with CH_4 ranging 10.5%–12.4%, C_2–C_4 ranging 27.5%–29.0%, and long-chain hydrocarbon (C_{5+}) of 59.0%–60.4%. The olefin/paraffin ratio in C_2–C_4 is in 1.14–1.24. In comparison, Fe@C-600-H_2 and unsupported-Fe catalyst exhibit inferior selectivity, namely higher methane selectivity (23.4% and 28.7%), higher C_2–C_4 selectivity (38.0% and 56.6%), lower C_{5+} selectivity (38.6% and 14.7%), and lower olefin/paraffin ratio in C_2–C_4 (0.95 and 0.40) at 24 h, respectively. Moreover, the stability of the catalysts can be investigated from the long-time run of FTS up to 196 h, as shown in Fig. 5. It can be observed that under the applied process condition, the CO conversion evolution with time on stream is exerted by pyrolysis conditions. The CO conversion of Fe@C-600 shows a gradual increase from 65% to 74.5% followed by a stable stage. Fe@C-600-H_2 also shows relatively stable activity. Other catalysts suffer a different degree of deactivation (with a deactivation rate of 13.1% for Fe@C-500, 20.1% for Fe@C-700 and 47.9% for unsupported-Fe in terms of CO conversion decline).

Because the aim of this work is to study the roles of θ-Fe_3C and carbon shell in FTS, it is very critical to evaluate the structure and phase during and after FTS reaction. It is worth noted that iron-based Fischer–Tropsch catalysts are easily oxidized after exposing to air and does not represent the actual catalyst phase anymore [33]. In this work, there are almost no iron oxides in the fresh catalyst (Fe@C-600, Fe@C-700, Fe@C-600-H_2 and unsupported-Fe) according to the XRD and MES results. It can be proved that the method of transferring and storing the catalysts in this work can effectively prevent the catalyst from being oxidized due to exposure to the air. Hence, the characterization results of the fresh catalysts and used catalysts can truly reflect the influence of the pyrolysis and reaction conditions on iron phase, without being disturbed by exposure to the air. The structure of catalysts after FTS was characterized by TEM as shown in Fig. 6. In Fe@C-500 as shown in Fig. 6(a), the particles without graphene shells aggregate apparently after FTS. In Fe@C-700 in Fig. 6(c), larger particles with obviously oxide phase can be observed. In unsupported-Fe catalysts, severe aggregation occurred as shown in Fig. 6(e). For the Fe@C-600 and Fe@C-600-H_2, uniform distribution of Fe, C, O is shown in Fig. 6(b), (d). Besides a small part of large particles, the structure of nanoparticles encapsulated in graphene shell is maintained after FTS. The θ-Fe_3C particles after FTS are still uniformly embedded in the carbon matrix, and the particle size does not change significantly compared with fresh catalysts. As for iron-containing phase, there are θ-Fe_3C, α-Fe, and iron oxide in the fresh catalysts according to XRD and MES. Iron oxide is considered inactive in FTS. The phase change of α-Fe is illustrated by characterization of the Fe@C-600-H_2 after FTS. From XRD pattern (Fig. S9) and MES spectra (Fig. S10), it can be observed that part of α-Fe evolved into ε-Fe_2C and χ-Fe_5C_2, which is consistent with previous reports [32,34]. However, as far as we know, the phase evolution of θ-Fe_3C in reaction conditions is rarely reported. In this work, take Fe@C-600 as an example to illustrate the phase transformation of θ-Fe_3C during reaction process. The iron phase and composition of Fe@C-600 after reaction for 50 h and 120 h were characterized by XRD and MES, respectively. From the XRD pattern (Fig. S11), diffraction line corresponding to θ-Fe_3C, α-Fe, and Graphite-carbon is observed in both spent catalysts. From the MES spectra in Fig. S12 and fitted parameters in Table S3, the catalyst after 50 h of reaction contains 88.3% θ-Fe_3C and 6.1% α-Fe. The catalyst after 120 h shows identical composition of 88.9% θ-Fe_3C and 5.7% α-Fe. α-Fe was always present during the reaction, which is different from the reported in the literature that α-Fe would transform into iron carbide or iron oxide [18,35]. In Fe@C-600, α-Fe may be wrapped inside θ-Fe_3C. The coated iron phase is difficult to contact with the reactant, therefore, there is no phase transformation [21,32]. Additionally, it is worth noting that θ-Fe_3C exhibits excellent stability in the reaction condition in this work. Therefore, in Fe@C-600, θ-Fe_3C plays major catalytic role.

It is accepted that the activity of iron-based catalysts in FTS is influenced by particle size, textural properties, and catalysts phase, etc. Although Fe@C-600-H_2 and unsupported-Fe are both composed of α-Fe, the FTY values of them are quite different, 203.0 and 39.5 $\mu mol_{CO} \cdot g_{Fe}^{-1} s^{-1}$, respectively. The larger iron particles and lower specific surface area of unsupported-Fe provides fewer active sites and therefore lower activity. Then the influence of phase on activity is described by comparing Fe@C-600 and Fe@C-600-H_2 since there is different phase composition but identical carbon shell in these two catalysts. It can be observed that θ-Fe_3C dominated Fe@C-600 exhibited lower CO conversion (65%) than that of Fe@C-600-H_2 (78.7%) composed of α-Fe. It is reasonable as theoretical calculation shows that θ-Fe_3C has a higher CO dissociation barrier of 1.92 eV than α-Fe of 1.53 eV [36,37]. The relatively weak CO dissociation ability of θ-Fe_3C may be the reason that Fe@C-600 possess lower CO conversion than Fe@C-600-H_2.

The carbon shell affects the stability of the catalyst. In general, small nanoparticles are prone to aggregate into larger clusters in the reaction process [38]. In this work, however, Fe@C-600 and Fe@C-600-H_2 possessing smaller particle size exhibit good resistance to particle aggregation. Although containing the same iron phase with Fe@C-600-H_2, unsupported-Fe without support underwent more severe aggregation and poor reaction stability. It can be

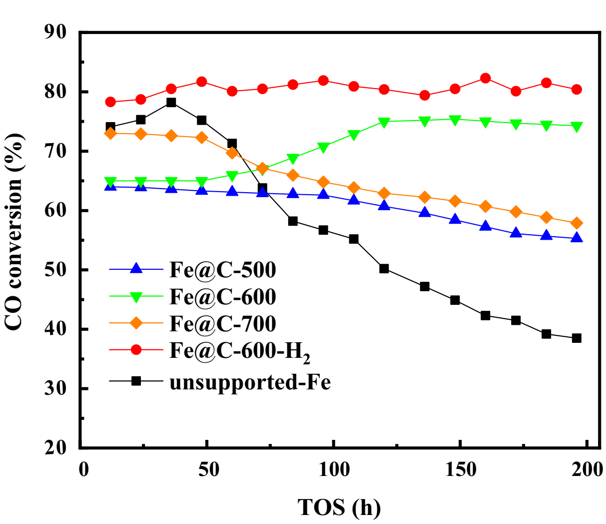

Fig. 5. CO conversion with time on stream.

Table 3
FTS performance of catalysts.

Catalysts		Fe@C-500[a]		Fe@C-600[a]		Fe@C-700[a]		Fe@C-600-H_2[a]		Unsupported-Fe[b]	
Time on stream (h)		24.0	184.0	24.0	184.0	24.0	184.0	24.0	184.0	24.0	184.0
CO conversion (%)		64.1	55.7	65.0	74.5	73.2	58.5	78.7	81.5	75.3	39.2
FTY ($\mu mol_{CO} \cdot g_{Fe}^{-1} s^{-1}$)		161.1	140.7	176.5	201.4	163.6	130.0	203.0	210.5	39.5	20.6
CO_2 selectivity (%)		35.2	36.7	33.3	33.7	33.9	34.2	37.3	38.6	38.5	43.66
HC distribution (wt%)[c]	CH_4	11.5	13.9	10.5	11.0	12.4	13.7	23.4	22.9	28.7	40.0
	C_2-C_4	29.0	35.1	28.1	30.4	27.5	30.1	38.0	36.9	56.6	57.6
	C_{5+}	59.0	50.1	60.4	57.9	59.6	56.1	38.6	40.2	14.7	2.4
Olefin/Paraffin (C_2-C_4)		1.14	1.15	1.20	1.25	1.24	1.21	0.95	1.01	0.40	0.35
α		0.68	0.61	0.69	0.68	0.67	0.64	0.49	0.50	0.30	0.21
Mass balance (wt%)		97.1	96.8	96.2	96.5	96.8	96.3	96.1	95.6	96.3	95.5

[a]Reaction conditions: $H_2/CO = 1.0$, 300 °C, 2.0 MPa, and a WHSV of 8000 ml/g/h.
[b]Reaction conditions:$H_2/CO = 1.0$, 300 °C, 2.0 MPa, and a WHSV of 3000 ml/g/h.
[c]Hydrocarbon distribution is calculated on total hydrocarbons.

inferred that the remarkable difference in stability is related to the carbon structure of catalysts. This can be further demonstrated that among the Fe@C-x catalysts, the graphene shell of Fe@C-600 can effectively protect the iron nanoparticle from aggregation during FTS reaction. In contrast, the amorphous carbon matrix cannot yield an adequate protective effect for particles in Fe@C-500. In Fe@C-700, large particles (40 nm) are mostly oxidized into Fe_3O_4. Probably, the thick graphite shell forms deep and narrow transmission channels, which is inefficient to diffuse water vapor from θ-Fe_3C surface and therefore the iron nanoparticles are oxidized. The volume expansion caused by oxidation destroys the surrounding graphite shell and further results in particle aggregation. In summary, the graphene shell of Fe@C-600 and Fe@C-600-H_2 protected the small particles from aggregation and stabilized the catalysts. In addition, it is worth noting that CO conversion of Fe@C-600 shows a gradual increase from 65% to 74.5%. It was discussed above that the phase composition of Fe@C-600 remains stable during reaction. Therefore, it is excludes that the increase of CO conversion rate between 50 h and 120 h is caused by phase transformation. One of the hypotheses for the activity increase is that the carbon fragments blocked in the pores inhibited the contact between the active site and the reactant. And as the reaction proceeds, the carbon fragments are removed, leading to the enhanced activity.

It has been reported that particle size, carbon support [5], CO conversion [39], and iron phase [10] could affect the selectivity. Compared with Fe@C-600-H_2, unsupported-Fe with larger particle size exhibited inferior product selectivity of 28.7% CH_4, 14.7% C_{5+} at 24 h, despite the same iron phase composition of the two catalysts. The θ-Fe_3C dominated Fe@C-x catalysts shows nearly identical product selectivity even with different carbon shell. Therefore, the effect of different carbon shell of MOFs-derived catalysts on selectivity could be ignored in this work. Herein, to illustrate the relationship of iron phase and product distribution, the selectivity of Fe@C-600 and Fe@C-600-H_2 is compared likewise under similar CO conversion levels (74%-81.5% at 184 h) and the same carbon shell. According to MES, Fe@C-600 contains 87.2% θ-Fe_3C and Fe@C-600-H_2 is composed of 100% α-Fe. After relating the selectivity with the iron phase content for the above two catalysts, it is found that θ-Fe_3C is related to lower CH_4 selectivity and higher C_{5+} selectivity, while α-Fe is the opposite. Thus, it is reasonable to conclude that θ-Fe_3C can indeed increase the selectivity of long-chain hydrocarbons and reduce the selectivity of methane. Generally, the less yield of CH_4 during FTS as a by-product accompanies with the higher yield of long-chain hydrocarbons, which is a crucial factor to evaluate the selectivity of FT catalysts. As α-Fe is usually transforms into ε′-$Fe_{2.2}C$ and χ-Fe_5C_2 during FTS reaction, it is important to identify the difference of various iron carbides for better understanding the selectivity difference of catalysts [4,40]. According to the calculated reaction energies and effective barriers, CH_4 formation is more favorable on Fe_5C_2 (010) (−0.23 and 1.54 eV) and Fe_2C (011) (−0.14 and 1.36 eV), while Fe_3C (001) (−0.18 and 1.98 eV) is inactive toward CH_4 formation.[36] Besides the theoretical calculation, Wezendonk et al reported that methane formation is minimized to 12% at 340 °C, 2 MPa when θ-Fe_3C is produced at 900 °C [41]. In order to further evaluate the chain growth capacity of catalysts, the chain-growth probability values (α) are shown in Table 3. The θ-Fe_3C dominated catalysts exhibited α of 0.67–0.69, which is relatively higher than the reported unpromoted Fe@C catalysts of 0.45–0.48 [15]. Moreover, it is well known that promoters such as potassium, sodium, and sulfur can effectively increase the FTS activity and the selectivity to long-chain hydrocarbon and olefins [4,42]. Adding alkali promoters to the active phase (θ-Fe_3C) with high chain growth ability may achieve higher long-chain hydrocarbon and olefins with less dosage, Therefore, the effect of alkali promoters on θ-Fe_3C is worthy of further study. However, it is reported that the further increase of the potassium content tends to notably enhance the deactivation rate, coinciding with the formation of inactive carbon species surrounding the catalysts surface which prevent the contact of catalysts surface with the gaseous reactants [42,43]. These above properties exhibited by alkaline additives may mask the intrinsic catalytic properties of the θ-Fe_3C. Hence, the role of θ-Fe_3C was investigated on unpromoted catalysts in this work.

θ-Fe_3C is considered to be a deactivation phase. For example, De Smit et al reported that the deactivation of the unsupported H_2 treated catalysts paralleled the occurrence of θ-Fe_3C, which may be due to the very small and amorphous clusters of θ-Fe_3C was generated [9]. Herranz et al. reported that θ-Fe_3C was less active in FTS because the formation of θ-Fe_3C was associated with the buildup of carbonaceous species that could occupy the active site [38,44]. In recent years, θ-Fe_3C is reported to exhibit relatively high activity in FTS. Zhang et al synthesized Fe@C core–shell catalysts with θ-Fe_3C phase by sol–gel approach.[45] A highest CO conversion of 98.5% was observed and the conversion could remain above 93.0% at 180 h time on stream at 300 °C. In this work, θ-Fe_3C dominated Fe@C-x catalysts were synthesized by ethyne assisted pyrolysis of MOFs precursor. And θ-Fe_3C is proved to be FTS active and exhibits a high capability to converting CO into long-chain hydrocarbons.

The carbon deposition is one of the reasons for deactivation of FTS catalysts during reaction. The inactive carbonaceous compounds including graphitic carbon, amorphous carbon and coke on the surface of the catalysts block active surface sites leading to lower activity and higher methane selectivity. Besides, pore blocking by carbonaceous deposits is reported as well, resulting in a drastic drop in accessible catalytic surface area [38]. Different from the dense carbonaceous compounds, loose and porous carbon

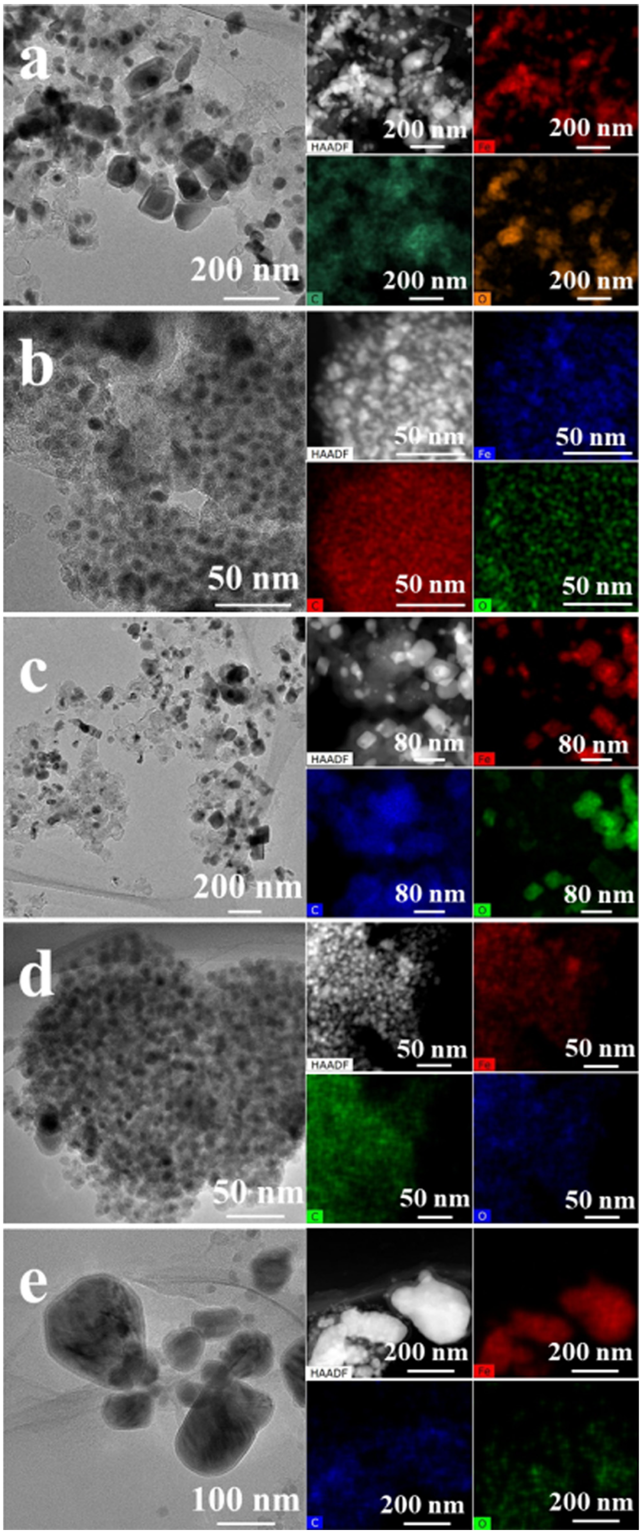

Fig. 6. TEM images of (a) Fe@C-500; (b) Fe@C-600; (c) Fe@C-700; (d) Fe@C-600-H_2; (e) unsupported-Fe and their HADDF and EDS mapping of Fe, C, O after FTS.

shell (amorphous carbon, graphene shell and graphite shell) around the particles is fabricated in a controllable way in this work. Amorphous carbon and thick graphite shells cannot protect particles from agglomeration. Interesting, the graphene shell can act as a partition to obtain smaller nanoparticles which possess higher active sites and prevent the aggregation of particles during reaction, effectively improve the stability of the catalysts. The suitable graphene shell can be widely used in other structure and size sensitive reactions to optimize the reaction performance.

4. Conclusion

In this work, MIL-101(Fe) pyrolyzed in C_2H_2 produces θ-Fe_3C dominated Fe@C catalysts with nanoparticles encapsulated uniformly in the porous carbon matrix. As the ethyne decompose violently with increasing temperature, amorphous carbon, graphene carbon shell and graphite carbon shell are fabricated between the nanoparticles and carbon matrix, respectively. The Fe@C catalysts exhibited C_{5+} selectivity of 57%–60% and CH_4 selectivity of 10%–12% at 300 °C, 2.0 MPa, and H_2/CO = 1.0. During the reaction, catalysts with graphene shell showed good stability without obvious particle aggregation.

During the MOFs pyrolysis in ethyne, iron species are reduced and carburized into α-Fe and/or θ-Fe_3C by carbon and hydrogen produced from MIL-101 and ethyne decomposition. The μ_c plays a dominant role in the formation of θ-Fe_3C, which is proportional to the content of θ-Fe_3C generated. Increasing μ_c is achieved by decomposing ethyne in higher temperature, therefore produce large amount θ-Fe_3C. Besides, catalyzed by iron, carbon fragments around iron particles condense into carbon shell and the degree of graphitization increase with pyrolysis temperature. Without the interference of promoters (K, Na, S, etc.), θ-Fe_3C is confirmed to be an active phase for FTS, which possesses high C_{5+} selectivity and low CH_4 selectivity compared with α-Fe or iron carbides evolved from α-Fe. Carbon shells exhibited different effects on the catalytic performance of catalyst. The porous graphene shell around the nanoparticles inhibited the particle aggregation and improved the stability of catalysts. The amorphous carbon shell and graphite shell cannot effectively protect the catalyst from oxidation and particle aggregation. Our research provides further understanding of the structure-performance relationship in iron-based FTS catalysts.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jcat.2020.11.033.

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